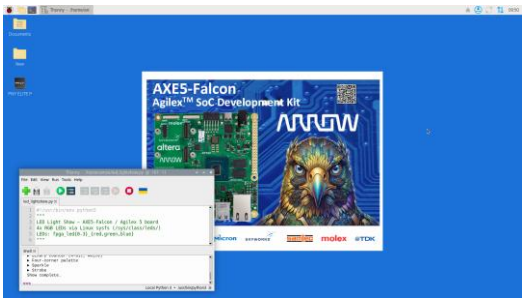
AXE5-Falcon



Desktop Linux Reference design

The Desktop Linux reference design brings a full Linux desktop to the Altera® Agilex™ 5 SoC on the Arrow AXE5-Falcon. Agilex 5 has no integrated GPU, so the Altera VVP Video Frame Reader (VFR) IP reads the Linux frame buffer from HPS DDR and drives 1920x1080p60 HDMI from the FPGA fabric.





A DRM/KMS platform driver presents the VFR to Linux as a standard display device, supporting Wayland, X11 and the frame buffer console. The XRGB8888 frame buffer lives in an 8 MB contiguous, non-cacheable CMA region, and page flips just update the VFR base address, so new frames latch without interrupting the video. The VFR runs at 200 MHz and the display pipeline at 148.5 MHz.

Graphics are rendered by the A55 Dual core processors. The driver supports Linux kernel 6.12 through 6.18. With it, the AXE5-Falcon boots straight to Linux on a standard HDMI monitor.

Partners:

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Key Features: Arrow AXE5-Falcon Kit

- A5EE013BB23BE6SCS
  - * 138 kLE
  - * Dual A55 SoC
  - * 376 multipliers 18x19
  - * 4.93 Peak INT8 (TOPS)
  - * 6.99 Mb (M20K) internal memory
  - * MIPI D-PHY interface
  - * LVDS pairs at 1.250 Mbps
  - * Secure Device Manager
- DDR4: 2x 1Gbit x16 (1Gb x 32)
- Dual Arm® Cortex®-A55 HPS running Linux (Raspberry Pi OS desktop))
- 1920x1080p60 HDMI output via the VVP Video Frame Reader
- DRM/KMS display driver: Wayland, X11 and frame buffer console
- 8 MB contiguous CMA framebuffer, XRGB8888 (32-bit)
- One USB-C cable for power, FPGA programming and HPS console

Ordering number: AXE5-Falcon
Technical details: GitHub page